

A Virtual Reality System for the Assessment of Patients with Lower Limb Rotational Abnormalities

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ABSTRACT

Rotational lower limb abnormalities cause patellar mal-tracking which impacts young patients. Repetitive patellar dislocation may require knee arthroplasty. Surgeons employ CT to identify rotational abnormalities and make surgical decisions. Recent studies also demonstrated that immersive 3D visualisation is preferred when examining 3D volumes of patient's data. We describe a prototype VR system that allows orthopaedic surgeons assess patients' lower limb anatomy in an immersive three-dimensional environment and simulate the effects of surgical procedures such as corrective osteotomies on a standalone VR headset. Preliminary results show an increased understanding of the patient's specific anatomy and predicted surgery outcomes.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction Paradigms—Virtual Reality, User Interfaces

1 INTRODUCTION

Orthopaedic rotational lower limb anomalies are prevalent, affecting 5.8 per 100,000 people, with a higher frequency in individuals aged 10–17. Misaligned joint axes (Fig. 1) may induce joint overloading, exercise discomfort, and early osteoarthritis. Rotational anomalies are diagnosed by assessing the complete limb's rotational profile using clinical examination and imaging, preferably computational tomography (CT). The severity of deformity and the patient's age determine whether surgical therapy is required [2].

Thanks to modern imaging technologies, the amount of data collected pre-surgery has increased massively. Virtual reality (VR) technologies could help visualise and interpret patient-specific data, improving our understanding of human anatomy. Several studies investigated the use of VR across different medical fields. For example, a patient's brain 3D model is used by neurosurgeons for clip pre-selection on aneurysms, and cardiologists are using 3D models to plan invasive coronary artery bypass surgeries for Kawasaki disease [4,6]. As standalone VR becomes inexpensive and widely available, magnetic resonance imaging (MRI) and CT can be converted into 3D models that can be then observed and manipulated stereoscopically, improving medical judgment, clinical training, and operation planning. Furthermore, planning complex surgical procedures is a team effort; it has been demonstrated that sharing virtual space during surgical planning can be beneficial [1].

In this project, we employ VR to assess patients' anatomy and visualise surgical outcomes for de-rotational corrective surgeries such as femoral, tibial, and tibial-tuberosity osteotomies. Our prototype VR system (OrthopedVR) enables creating virtual incision points to "break" the bone and modify the rotational profile in each bone segment. This involves calculating new segment angles and

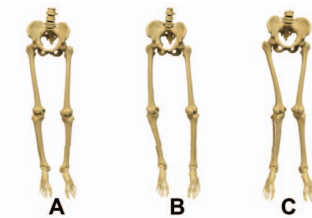


Figure 1: Difference between a *healthy* anatomy of the lower limbs (A), pathological anatomy affected by *varus deformity* (B), and pathological anatomy affected by *valgus deformity* (C).

visualising the angle change compared to the original by integrating an application-specific 3DUI which supports collaborative VR.

2 BACKGROUND

Lower-limb deformities are common orthopaedic issues. Soft connective tissues around the knee, adequate osseous volume, and appropriately aligned lower limb, contribute to a healthy patellofemoral joint. The rotational alignment of the femur and tibia determines patellofemoral tracking and knee joint function. The femoral version, tibial rotation, and knee joint rotation angle are of particular importance to orthopaedic surgeons [7]. Significant femoral anteversion causes an internally rotated gait unless an external tibial torsion pushes the leg outward to maintain a correct foot progression angle. Increased femoral anteversion and tibial torsion cause patellofemoral instability. Patellar mal-tracking causes instability and creates discomfort in the knee region. The first abnormality that impacts patella tracking is tibial valgus (knock knee syndrome) or varus (bow leg syndrome). Valgus tibias are common in pediatric patients. In certain cases, rotational abnormalities can be caused by a rotated distal femur [3].

The potential of VR surgical planning in clip pre-selection in aneurysms [4] has been demonstrated. According to the study, VR provided neurosurgeons with an improved spatial understanding of vascular anatomy. The success of their study was quantified by the outstanding rate of aneurysm closures. VR aided surgeons with planning oncologic liver surgeries [5], and was shown to curtail the number of procedure-related complications.

3 SYSTEM OVERVIEW & IMPLEMENTATION

Our simulation solution for de-rotational osteotomies incorporates a collaborative VR environment, a hybrid 3DUI, and an osteotomy simulation, all facilitated by optimisations for standalone VR headsets. We used Unity3D and deployed it on Meta Quest 2 headsets, enabling surgeons practice osteotomies using real patient data and visualise the results of the corrective procedure. Angle correction amounts may be assessed and visualised before operating. A surgeon initially marks the incision plane. Our algorithm then recalculates the geometry of the segmented bones and creates two superimposed bone segments that may be modified individually and fused back collaboratively, following the necessary angle calculations and modifications.

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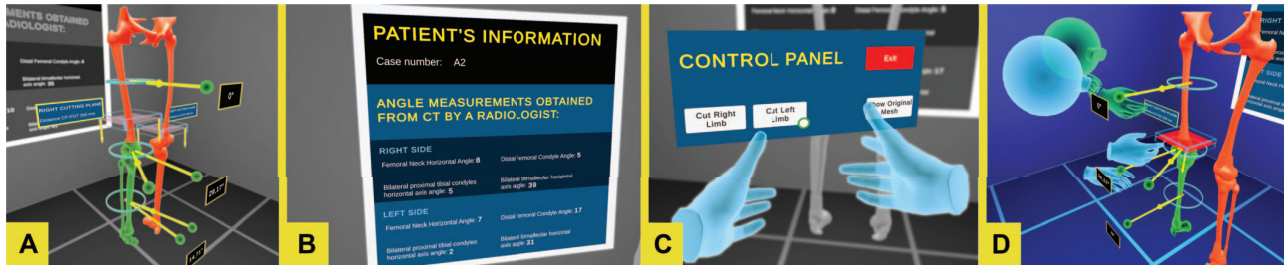


Figure 2: (A) The contextual 3DUI providing bone-cutting actions and relevant information visualisation. (B) The world-positioned UI, omni-present, communicating essential information. (C) Dynamic control panel UI, attached to the one hand of the user and controlled with their other hand, containing context-specific actions that get dynamically updated. (D) Surgeons collaborating on a patient's case.

3DUI for derotational osteotomies. We achieve simulated derotational osteotomies involving an application-specific 3DUI for decision making and information visualisation. We use contextual, world-positioned, and dynamic control panels (Fig. 2). Combining three UI design paradigms enables users to make treatment-related decisions and take actions with all necessary information readily available. Depending on the state, the UI keeps most items/options accessible and available to select/view. Only two physical buttons Index Triggers and Hand Triggers control our 3DUI.

Contextual UI. The lower limb segmentation is interacted with a contextual UI. The UI appears during segmentation, displaying a protractor (Fig. 2A). The user can rotate a lower limb bone-fragment by touching it and then rotating their wrist. Each bone segment's rotation angle is shown in the protractor. When altering bone structures, this UI provides relevant decision-making info.

World-positioned UI. The World-positioned UI presents critical patient/procedure data that should always be available and were gathered during routine clinical examination processes, e.g., lower limb rotational angles. This UI is always-present, avoiding users needlessly search the virtual environment for critical information.

Dynamic control panel. This is a point-and-click control panel, dynamically updated, visible at all times, attached to the left controller (user's left hand - for right-handed users; can be swapped for the right hand, for left-handed users) providing immediate access to the most essential functions. Such a point-and-click UI, circumvents the use of several physical controller buttons to perform the large variety of tasks required during pre-operative planning.

Collaborative osteotomies. The Photon Engine PUN2 solution from Exit Games GmbH (Germany) was incorporated in our implementation enabling VR client interaction. Remote procedure calls distribute script execution among all network users, synchronising position and rotation for all interactable items in the scene. Following initiation of the cut after positioning the cutting plane, a sequence of procedures create a new mesh segmentation with split-off portions. The 3D position and orientation of the cutting plane are used to construct a virtual slicing plane. We then exclude crossing vertices in the vertex list to separate the meshes. Two vertex lists are constructed at the intersection plane, and the final original vertices' sides of the cut are checked. Two meshes are then constructed from each vertex list. For proper illumination, vertex normals, bounds, and tangents are adjusted. Segments use orange and green materials; green materials represent user-manipulatable objects.

4 EVALUATION AND CONCLUSION

We evaluated surgeon performance using standard DICOM viewers and OrthopedVR, on five cases. Quantitative (task completion time) and qualitative (questionnaires) assessment demonstrated that immersive surgical planning in VR increases comprehension of structural anomalies, pathological anatomy, surgical accuracy, and clinical outcomes at a fraction of the time (Fig. 3) compared to DI-

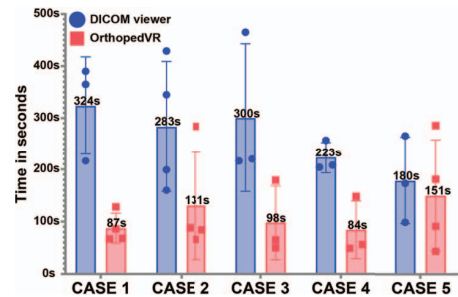


Figure 3: Task completion time: DICOM viewer versus OrthopedVR.

COM viewing, thus requiring less cognitive resources. OrthopedVR requires no modifications to existing clinical screening protocols, no additional data or further radiation exposure. To our knowledge, this is the first system to improve surgical planning for lower limb orthopaedic procedures, in VR.

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